⊛. USER BASED COLLABARATIVE FILTERING

Dataset:

	Item-1	Item-2	Item-3	Item-4
User A	4	3	5	?
User B	5	5	4	3
User C	4	3	5	3
User D	2	4	2	3.

Core logic:- Users similar to User A liked item 4, so User A will probably like it too.

step 1:- Calculate the similarities between User A and other users.

We ignore item 4 for now because User A hasn't rated it. We compare ratings for items 1, 2, 3.

Now the core Idea is assuming User A is similar to other users predict the rating for Item 4 which will be given by User-A.

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→ let's use Cosine similarity:

Formula:-

$$Sim(A,B) = \frac{\Sigma(Rating_A \times Rating_B)}{\sqrt{\Sigma Rating_A^2} \times \sqrt{\Sigma Rating_B^2}}$$

let's calculate (User A similarity with User B, User C, User D.

→ similarity (User A, User B)

Num :- $\Sigma(Rating_A \times Rating_B)$

=) Dot product : $(4 \times 5) + (3 \times 5) + (5 \times 4)$

$20 + 15 + 20$

$55.$

Den :- $\sqrt{\Sigma Rating_A^2} : \sqrt{4^2 + 3^2 + 5^2} = 7.07$

$\sqrt{\Sigma Rating_A^2}$

$\sqrt{\Sigma Rating_B^2} : \sqrt{5^2 + 5^2 + 4^2} = 8.12.$

$$Sim(A,B) :- \frac{55}{(7.07 \times 8.12)} = 0.958.$$

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similarly calculate (User A → User C)

(User A → User D).

User A, User C => 1.0 (perfect match)

User A, User D => 0.866.

Step 2:- Select Top Neighbors.

User A → User B – 0.958

User A → User C – 1

User A → User D – 0.866

let's choose top 2 neighbors based on similarity score which are User B & User C. to predict User A item 4 rating.

Step 3:- predict the Rating.

We take the weighted average of the rating given to item 4 by neighbor B & C.

$$\text{prediction} : \frac{\Sigma (\text{similarity score} \times \text{Rating})}{\Sigma \text{similarity}}.$$

$$\text{prediction}: \frac{(0.958 \times 3) + (1 \times 3)}{0.958 + 1.0}$$

prediction:- **3.0.**

Prediction for User A on item 4 is 3.

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